$$\frac{dy}{dx}+2xy-4x^3 = 4x-2xCe^{-x^2}+2x\left[2\left(x^2-1\right)+Ce^{-x^2}\right]-4x^3 = 0$$

Therefore, the given function is a solution of the differential equation $\frac{dy}{dx}+2xy-4x^3=0$. ■

Example 10.10

Show that $y=a\cos(\log x)+b\sin\left(\log x\right), x>0$ is a solution of the differential equation $x^2y''+xy'+y=0$.

Solution

The given function is $y=a\cos(\log x)+b\sin\left(\log x\right)$ ...(1)

where a,b are two arbitrary constants. In order to eliminate the two arbitrary constants, we have to differentiate the given function two times successively.

Differentiating equation (1) with respect to x, we get

$$y'=-a\sin\left(\log x\right)\cdot\frac{1}{x}+b\cos\left(\log x\right)\cdot\frac{1}{x}\Rightarrow xy'=-a\sin\left(\log x\right)+b\cos\left(\log x\right).$$

Again differentiating this with respect to x, we get

$$xy''+y'=-a\cos\left(\log x\right)\cdot\frac{1}{x}-b\sin\left(\log x\right)\cdot\frac{1}{x}\Rightarrow x^2y''+xy'+y=0.$$

Therefore, $y=a\cos(\log x)+b\sin\left(\log x\right)$ is a solution of the given differential equation. ■

EXERCISE 10.4

1. Show that each of the following expressions is a solution of the corresponding given differential equation.
   (i) $y=2x^2$; $xy'=2y$
   (ii) $y=ae^x+be^{-x}$; $y''-y=0$
2. Find value of m so that the function $y=e^{mx}$ is a solution of the given differential equation.
   (i) $y'+2y=0$ (ii) $y''-5y'+6y=0$
3. The slope of the tangent to the curve at any point is the reciprocal of four times the ordinate at that point. The curve passes through (2,5). Find the equation of the curve.
4. Show that $y=e^{-x}+mx+n$ is a solution of the differential equation $e^x\left(\frac{d^2y}{dx^2}\right)-1=0$.
5. Show that $y=ax+\frac{b}{x}, x\neq 0$ is a solution of the differential equation $x^2y''+xy'-y=0$.
6. Show that $y=ae^{-3x}+b$, where a and b are arbitrary constants, is a solution of the differential equation $\frac{d^2y}{dx^2}+3\frac{dy}{dx}=0$.
7. Show that the differential equation representing the family of curves $y^2=2a\left(x+a^{\frac{2}{3}}\right)$, where a is a positive parameter, is $\left(y^2-2xy\frac{dy}{dx}\right)^3=8\left(y\frac{dy}{dx}\right)^5$.

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